

Aviv Shaked

Cybersecurity Developer & Researcher · Haifa, Israel

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SUMMARY

C++20 cybersecurity developer and researcher specializing in network security, binary exploitation, and reverse engineering. Graduate of **Magshimim+**, Israel's elite national cybersecurity track, consistently self-driven beyond the curriculum. Co-developed **HitchHiker**, a production covert communication platform embedding encrypted messages into live traffic with zero new packets; placed **2nd globally** in AI4Y (2023).

TECHNICAL SKILLS

Languages	C++20 · Python · C · x86-64 Assembly · SQL · Bash · C#
C++ & Systems	RAII · smart pointers · multi-threading · socket programming (TCP/UDP/raw) · CMake · modular builds
Security	Binary exploitation · ROP chains · heap exploitation · format strings · ASLR bypass · reverse engineering · network steganography · threat modeling · HackTheBox (RE, Pwn, Web, Crypto, Forensics)
Networking	L2-L7 stack · TCP/IP · DNS · TLS · ICMP · ARP · DHCP · bit-level field manipulation · kernel packet interception (NFQueue) · iptables · Wireshark · Scapy
Cryptography	ECDH · ChaCha20 · AES · Argon2id · HKDF · SHA-256 · key exchange · key derivation · hash attacks · applied in production via Crypto++
AI / ML	Neural networks · minimax search · value function approximation · PyTorch · Scikit-learn · LLM/API integration
Tools	Binary Ninja · IDA Pro · Ghidra · GDB · Wireshark · nmap · Burp Suite · Docker · Git · Kali Linux · AWS EC2 · JetBrains

PROJECTS

HitchHiker: Network Steganography Platform C++20 · 2025-2026 · Completed
Co-developed with Daniel Varkin · Source available on request · Architecture: patcholie.github.io

C++20 Crypto++ NFQueue FTXUI Redis SQLite Docker AWS EC2 CMake

- Co-architected a covert communication platform that embeds encrypted messages into live network traffic: **zero new packets generated**, no observable network presence, undetectable by design
- Full cryptography stack (key exchange, symmetric encryption, key derivation, password hashing) applied at bit-field precision across multiple protocol layers via kernel-level packet interception
- Pluggable carrier architecture** behind a unified HitchHikerAPI; three independent CMake targets (lib / server / client), Redis + SQLite storage, full TUI (FTXUI); deployed on AWS EC2 via Docker

Multiplayer Trivia Game System C++ / C# / Unity · Magshimim · 2024
C++ C# Unity SQLite TCP Sockets

- Designed a custom application-layer protocol from scratch; concurrent TCP server, C# Unity client, and SQLite database integrated end-to-end with clean OOP architecture; scored 130/100 including bonus marks

Analysis: AI Wound Treatment Platform Python · 2023 · Public

- Co-led development of an AI-powered wound diagnosis and treatment guidance platform; **placed 2nd globally** in the AI4Y World Competition (2023) across international teams

Mancala-AI Python · PyTorch · 2025-2026

- Multi-agent AI system for Mancala with four agents (Random, Greedy, Heuristic, NN); NN uses 2-ply minimax with a trained MLP value function achieving 94% win rate vs Heuristic; full training pipeline, tournament evaluator, and GUI

EXPERIENCE

Python Automation Developer, AllBikes 2022-2023 · Remote

- Built Python automation tools for web scraping and product catalog structuring; **tooling adopted by the full team**, delivered independently in a remote professional environment while in secondary school

EDUCATION

Magshimim Cyber, Advanced Track (Magshimim+) 2023-2026
Israel's elite national cybersecurity program, selective advanced track. Curriculum: networking, binary exploitation, reverse engineering, cryptography, systems programming (C/C++), Linux, team-based security development

BOSMAT, Haifa 2020-2026
Computer Science and AI, 10 study units (maximum track)

AWARDS & RECOGNITION

2nd Place, AI4Y World Competition · 2023 · Global
Magshimim+ Advanced Track · Graduate 2026 Magshimim Cyber · Graduate 2026